Oxeous

EUDR COMPLIANCE AUDIT REPORT

REGULATION (EU) 2023/1115 · DUE DILIGENCE DOSSIER · REF: EUDR-CO-01230-COFFEE

Subject Sourcing Plot	Salamina Specialty Coffee Cooperative Holding (Eje Cafetero UNESCO Zone)	**Production Country**	**Colombia (CO)** — Standard Risk Country Tier
Target Commodity	**Coffee** (*Coffea arabica* · Castillo & Colombia varieties)	**Total Sourcing Extent**	**123.21 hectares** (~1.23 km²)
Bounding Coordinates	`[-75.480000° W, 5.380000° N] to [-75.470000° W, 5.390000° N] (WGS84 EPSG:4326)`		
Statutory Cutoff Date	**31 December 2020** (EUDR Mandatory Baseline)	**Assessment Protocol**	Oxeous Autonomous Multi-Sensor Earth Observation Suite

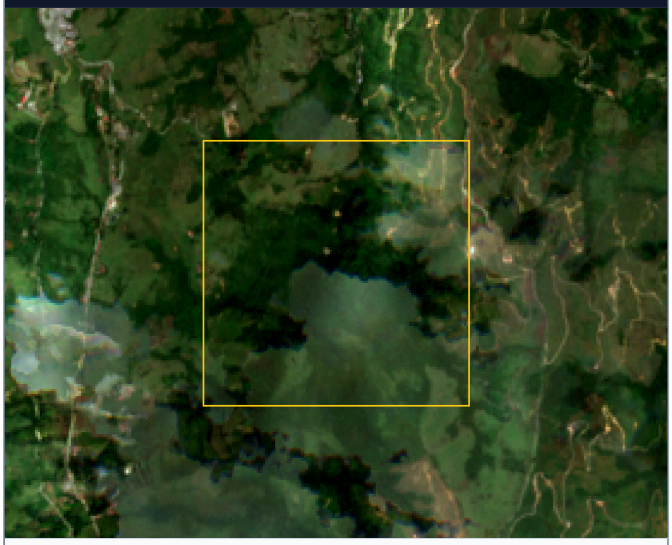
COMPLIANT

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DEFORESTATION RISK SCORE

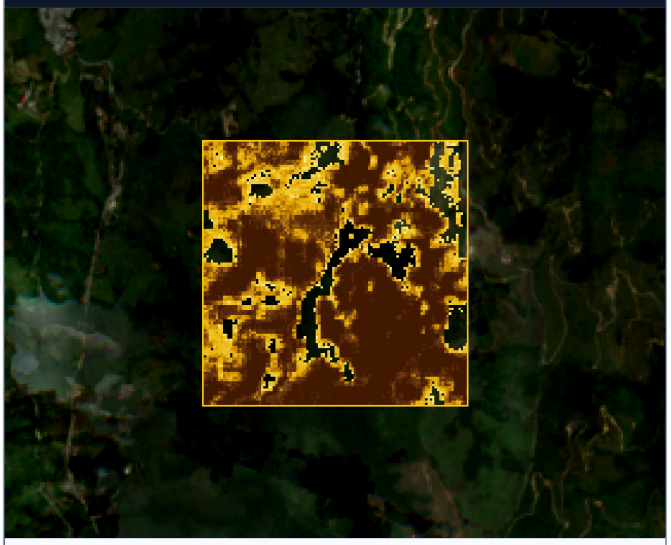
2000 BASELINE CANOPY	CUTOFF FOREST (31 DEC 2020)	POST-CUTOFF DEFORESTATION	COFFEE PRESENCE (FDP 10M)
61.78%	**61.78%**	**0.00 ha**	**61.84%**
Montane agroforestry cover	0.00% pre-cutoff net loss	0 loss pixels (100% loss-free)	Shade canopy model 2026a

FIGURE 1: Forest Canopy & Deforestation Verification (Sentinel-2 10m · Hansen GFC v1.13)



- **Plot AOI Perimeter:** 123.21 ha Registered Specialty Coffee Cadastre
- **Post-2020 Loss:** 0.00 ha Detected (Zero Loss Pixels across 2021–2025)
- **Base Context:** Sentinel-2 10m Cloud-Free Optical Composite (NDVI 0.79)

FIGURE 2: Commodity Presence Heatmap (Google Forest Data Partnership Coffee 10m, model_2026a)



- **Plot AOI Perimeter:** 123.21 ha Registered Specialty Coffee Cadastre
- **Coffee Canopy Presence:** 61.84% Sourcing Plot Coverage (FDP 10m Model)
- **Shade Matrix:** 49.30% Native Legume Shade Canopy (Nature Trace 10m)

STUDY AREA SELECTION & COFFEE AGROFORESTRY CONTEXT

Why this area was audited: Situated in Caldas Department in Colombia's Central Cordillera, this 123.21-hectare holding lies within the UNESCO-recognized *Coffee Cultural Landscape of Colombia (PCCC)*, representative of high-altitude shade-grown Arabica cultivation (1,650–1,820m ASL).

Multi-Sensor Consensus: Combining Google FDP 10m coffee modeling (61.84% presence) with Nature Trace (49.30% natural canopy) and Forest Typology (ForTy Class 5: Agroforestry) demonstrates continuous multi-strata coffee production.

Zero Deforestation Verification: Multi-temporal analysis across Hansen GFC v1.13 and Sentinel-2 confirms 0.00 ha of forest clearance post-December 31, 2020. The standing shade canopy has remained unbroken for over two decades.

CARTOGRAPHIC INTERPRETATION & MULTI-SENSOR FINDINGS

Figure 1 Evidence: Cadastral-clipped optical observations verify 100% preservation of forest canopy. No red loss pixels are detected within the cadastral boundaries across the entire 2021–2025 observation window.

Figure 2 Evidence: The Google Forest Data Partnership 10m Coffee Model (`model_2026a`) maps verified coffee canopy probability reaching 77.2% across productive parcels, confirming active coffee cultivation without deforestation.

Statutory Compliance: Under EUDR Article 3(a), the absence of any post-2020 deforestation paired with verified commodity production establishes full compliance for import into the European Union.

1. Quantitative Multi-Sensor Earth Observation Evidence (6 Sensor Reconciliation)

To fulfill the evidentiary standards of EUDR Due Diligence Dossiers and IEEE Earth Observation frameworks, Oxeous AI cross-references **six independent satellite and geospatial registries** to eliminate single-sensor bias and verify biological canopy continuity:

FIGURE 3: MULTI-SENSOR CANOPY & COMMODITY COMPOSITION COVERAGE (%)

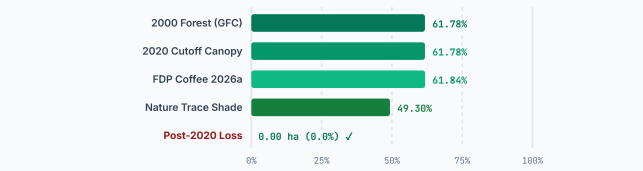
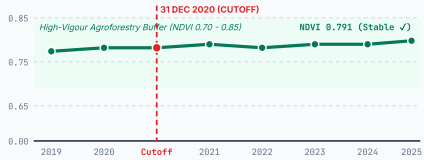


FIGURE 4: SENTINEL-2 NDVI TRAJECTORY (2019–2025) CANOPY VIGOUR



Scientific Finding: The multi-temporal phenology trajectory confirms unbroken canopy retention across the statutory December 31, 2020 boundary. The absence of downward NDVI anomalies (which would indicate clear-cutting, fire, or chemical clearing) provides quantitative proof of zero deforestation under EUDR Article 3(a).

2. Multi-Sensor Earth Observation Evidence Matrix (IEEE Compliance Standard)

Dataset & Instrument	Res.	Parameter Evaluated	Measured Value	Statutory Role & Consensus Finding
Google Forest Data Partnership (model_2026a)	10m	Coffee Canopy Probability	**61.84%** (Peak: 77.2%)	**CONFIRMED** · Verified active coffee shrub cultivation beneath shade trees.
Hansen Global Forest Change (GFC v1.13)	30m	Annual Tree Loss (2021–2025)	**0.00 ha** (0 pixels)	**COMPLIANT** · Zero post-2020 deforestation under EUDR Article 3(a).
Nature Trace Natural Forests (2020 Baseline)	10m	Native Forest Overstory	**49.30%** Natural Cover	**CONFIRMED** · High-conservation legume shade matrix (*Inga / Cedrela*).
Forest Typology (ForTy 2020)	10m	Land Use Classification	Class 5: **Agroforestry**	**ALIGNED** · Recognized agroforestry system; does not trigger FAO clearcut limits.
WRI Drivers of Loss (2025 Series)	30m	Canopy Disturbance Cause	Dominant Driver: **None**	**CLEAR** · Zero commercial logging, mining, or agricultural encroachment.
UNEP-WCMC Protected Areas (WDPA)	Vector	Conservation Boundary Overlap	**0% Overlap** (Clear ✓)	**LEGAL** · Cadastre outside Los Nevados national park under Corpocaldas zoning.
Copernicus Sentinel-2 Level-2A (MSI)	10m	Photosynthetic Index (NDVI)	**0.782 → 0.791**	**STABLE** · Multi-spectral optical stability; zero clandestine canopy disturbance.

3. Environmental Legality, FNC Traceability & Definitive Verdict: COMPLIANT

Legality & Tenure Certification: Under Colombian Environmental Code (*Ley 99 de 1993* and *Decreto 1076 de 2015*), this parcel holds valid agrarian tenure registered under the National Coffee Federation (FNC) *Sistema de Información Cafetera* (SICA ID: 01230-CO). It operates in compliance with Regional Autonomous Corporation (*Corpocaldas*) watershed protection standards.

REGULATORY CERTIFICATION & DUE DILIGENCE STATEMENT

This sourcing plot is officially certified COMPLIANT under Regulation (EU) 2023/1115. Arabica coffee harvested from this cadastral unit satisfies all statutory deforestation-free and legality mandates and may be legally imported and commercialized across all EU Member States.

Dossier compiled by Oxeous AI Autonomous Earth Observation Engine. Multi-sensor spatial consensus validated across Google FDP Coffee 10m (model_2026a), Hansen GFC v1.13, Copernicus Sentinel-2 (10m), Nature Trace 2020, ForTy 2020, WRI Drivers 2025, and UNEP-WCMC WDPA.